Discrete Applied Mathematics 213 (2016) 101–113

Contents lists available at ScienceDirect

Discrete Applied Mathematics

journal homepage: www.elsevier.com/locate/dam

Partitioning de Bruijn graphs into fixed-length cycles for robot identification and tracking

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ARTICLE INFO

Article history:
Received 8 November 2015
Received in revised form 6 April 2016
Accepted 10 May 2016
Available online 21 June 2016

Keywords:
Graph theory
Robot network
de Bruijn graph
Graph decomposition
Pose estimation
Linear feedback shift register

ABSTRACT

We propose a new camera-based method of robot identification, tracking and orientation estimation. The system utilises coloured lights mounted in a circle around each robot to create unique colour sequences that are observed by a camera. The number of robots that can be uniquely identified is limited by the number of colours available, q, the number of lights on each robot, k, and the number of consecutive lights the camera can see, ℓ. For a given set of parameters, we would like to maximise the number of robots that we can use. We model this as a combinatorial problem and show that it is equivalent to finding the maximum number of disjoint k-cycles in the de Bruijn graph $\mathrm{dB}(q, \ell)$.

We provide several existence results that give the maximum number of cycles in $\mathrm{dB}(q, \ell)$ in various cases. For example, we give an optimal solution when $k = q^{\ell-1}$. Another construction yields many cycles in larger de Bruijn graphs using cycles from smaller de Bruijn graphs: if $\mathrm{dB}(q, \ell)$ can be partitioned into k-cycles, then $\mathrm{dB}(q, t\ell)$ can be partitioned into tk-cycles for any divisor t of k. The methods used are based on finite field algebra and the combinatorics of words.

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1. Introduction

A *robot network* is a collection of robots working together to achieve a common goal. In order for the robots in such a network to cooperate effectively, the ability to observe each other's movements is critical. In many applications, distinguishing between the robots is necessary, but is usually difficult because the robots are identical.

For example, in a *formation control* system, robots collectively arrange themselves in some fixed geometric configuration [2,9]. Each robot controls its position relative to its neighbours. To achieve this, the robot must continuously measure the position and determine the identity of each neighbour. Some formation control systems may also benefit from knowledge of the relative orientation of its neighbours, since this information can be used to coordinate views and improve the stability of the system.

We present a novel camera-based method for robot identification, orientation estimation, and approximate distance/angle measurements. The system uses a camera to observe sequences of coloured lights mounted on the robots. The lights are mounted in a circle around each robot (in a plane parallel to the ground), such that a camera may see only some of the lights. The sequences of colours are chosen so that any consecutive subsequence of sufficient length corresponds uniquely to a particular robot in a particular orientation.

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http://dx.doi.org/10.1016/j.dam.2016.05.013
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Fig. 1. An eBug displaying a colour sequence on eight of its sixteen LEDs. The camera viewing the eBug can only see three of these LEDs.

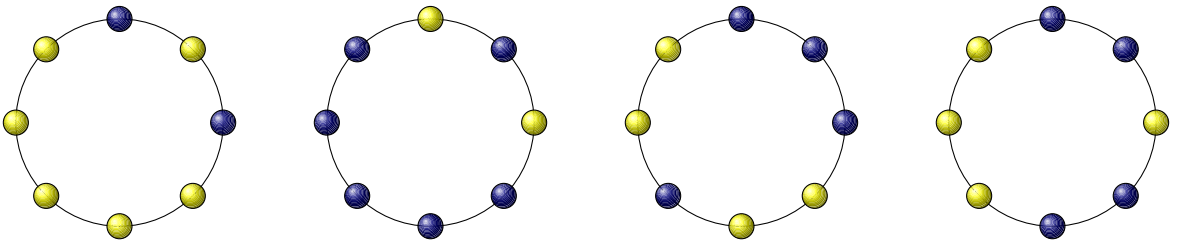


Fig. 2. Example eBug colouring with $q = 2$, $k = 8$ and $\ell = 5$.

This system was implemented in an existing network of *eBugs*. The *eBug* [8] is a robotics platform designed at Monash University's Wireless Sensor and Robot Networks Laboratory [28]. It is equipped with sixteen RGB LEDs (red, green and blue light-emitting diodes) on its perimeter, which can be programmed to display a sequence of colours. A photo of an eBug may be seen in Fig. 1.

Fig. 2 shows an example of a colouring of four eBugs. These eBugs have only eight LEDs, and use only two different colours for illustrative purposes. Any subsequence of five LEDs is coloured with a unique pattern. For example, the sequence ●●●○○ appears (counter-clockwise) only on the right side of the second eBug.

In a real system, there are limits on the number of colours a camera may reliably distinguish. Similarly, spatial resolution of the camera limits the number of detectable LEDs around each eBug. Therefore, for a given set of parameters, we want to maximise the number of eBugs that we can use in the system. This maximum, the *eBug number*, is formally defined below.

Definition 1 (*eBug Number*)**.** Suppose every eBug has k LEDs, each of which can be illuminated in one of q colours, and suppose further that a camera can reliably detect $\ell \leq k$ consecutive LEDs. An assignment of colours to the LEDs of all eBugs is *ℓ-valid* if the camera can distinguish each eBug in each of the k orientations. The *eBug number* $\mathcal{E}(q, k, \ell)$ is the maximum number of eBugs for which there exists an ℓ-valid assignment of colours.

As well as modelling an actual problem that arises in robot networks, determining eBug numbers is a natural combinatorial problem of independent interest.

Each of the q^ℓ possible sequences the camera can see cannot appear more than once, and each eBug uses k distinct sequences. This gives the following upper bound for the eBug number:

$$\mathcal{E}(q, k, \ell) \leq \frac{q^\ell}{k}. \tag{1}$$

Colourings that achieve the upper bound in (1) are called *optimal*. In such colourings, each sequence of ℓ colours appears on some eBug. Note that when $k = \ell$, no ℓ-sequence of a constant colour can appear on an eBug since all orientations would be identical. Thus optimal colourings can only exist for $k > \ell$.

A lower bound may be obtained by applying the Lovász local lemma [12,23]: consider a random colouring of n eBugs, with each of the nk LEDs coloured independently and uniformly at random. For each pair (i, j) of LED sequences (of length ℓ), let A_{ij} be the event that the same colour sequence has been assigned to i and j. Thus the colouring is ℓ-valid exactly when none of the events A_{ij} occurs. Since there are exactly nk LED sequences, and each sequence overlaps with at most $2\ell - 1$ other sequences, each event A_{ij} depends on at most $nk(2\ell - 1) - 1$ other events. The probability of each A_{ij} is at most $q^{-\ell}$ (less if i and j are overlapping). Therefore, by the local lemma, there is an ℓ-valid colouring whenever $eq^{-\ell}nk(2\ell - 1) \leq 1$, where e is Euler's number. Hence we obtain the following lower bound:

$$\mathcal{E}(q, k, \ell) \geq \left\lfloor \frac{q^\ell}{(2\ell - 1)ek} \right\rfloor.$$

For a fixed value of ℓ, this bound is within a constant factor of the upper bound in (1). In actual camera systems, however, it is reasonable to assume that ℓ is proportional to k, since a camera can usually detect a fixed arc of the LED circle. Thus the lower bound is rather crude, and ultimately we would like to solve the following problem.

Problem 1. Determine $\mathcal{E}(q, k, \ell)$ exactly.

For small values of q and ℓ, a computer search was performed to find large ℓ-valid colourings. Surprisingly, optimal colourings were found in many cases. These experiments confirm the following conjecture for all q and ℓ with $q^\ell \leq 81$. While Problem 1 is likely to be very difficult to solve in general, a mathematically interesting problem is to characterise when optimal colourings exist (hopefully by proving Conjecture 1).

Conjecture 1. $\mathcal{E}(q, k, \ell) = \frac{q^\ell}{k}$ *whenever* k *divides* q^ℓ *and* $k > \ell$.

This paper provides constructions for some infinite families of optimal colourings, and as such gives evidence to support this conjecture.

In Section 2, Problem 1 is shown to be equivalent to finding many cycles in a *de Bruijn graph*, with Conjecture 1 corresponding to a partition into cycles (see Proposition 1). Existing results about *de Bruijn sequences* are also discussed.

A well-known algebraic construction of de Bruijn sequences is given in Section 3; we extend this construction to prove some existence results for eBug colourings. The major result of this section is Theorem 1, which proves Conjecture 1 for infinitely many values.

Theorem 1. *If* q *is a prime power and* $\ell \geq 1$, *then* $\mathcal{E}(q, q^{\ell-1}, \ell) = q$.

In Section 4, we introduce *necklaces* and how they relate to de Bruijn graphs. We then prove the following theorems, both of which yield large eBug colourings from smaller ones.

Theorem 2. *Fix a value of* ℓ *and set* $\mathcal{E}_1 = \mathcal{E}(q_1, k_1, \ell)$ *and* $\mathcal{E}_2 = \mathcal{E}(q_2, k_2, \ell)$. *Then*

$$\mathcal{E}(q_1q_2, \operatorname{lcm}(k_1, k_2), \ell) \geq \gcd(k_1, k_2)\, \mathcal{E}_1\, \mathcal{E}_2.$$

Theorem 3. $\mathcal{E}(q, tk, t\ell) \geq \frac{k^{t-1}}{t}\mathcal{E}(q, k, \ell)^t$ *whenever* t *divides* k.

These theorems preserve optimality, so we may use them to find optimal colourings for large numbers of eBugs.

2. Preliminaries

2.1. de Bruijn graphs

A valid colouring of eBugs has an interesting interpretation as cycles in a de Bruijn graph. These graphs were discovered independently by de Bruijn [10] and Good [15] in 1946.

Definition 2. The ℓ-th order q-ary *de Bruijn graph* $\mathrm{dB}(q, \ell)$ is the directed graph with vertex set $V = \mathbb{Z}_q^\ell$ and edge set $E = \{(a_0a_1 \ldots a_{\ell-1}, a_1a_2 \ldots a_\ell) \mid a_i \in \mathbb{Z}_q\}$.

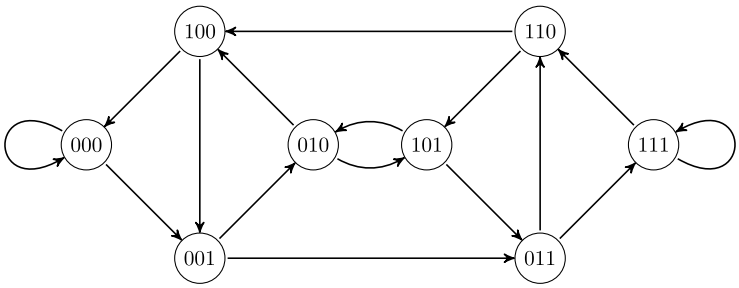


Fig. 3. Example de Bruijn graph—dB(2, 3).

The vertices of dB(q, ℓ) are words of length ℓ over an alphabet of size q. There is an edge from u to v if shifting u left and appending any letter gives v. An example of such a graph is shown in Fig. 3.

There is also an alternative, equivalent definition of de Bruijn graphs that involves iteratively taking line digraphs [29]. In this construction, dB(q, 1) is defined as the complete digraph on q vertices with loops. Higher order de Bruijn graphs are defined as follows: dB(q, $\ell + 1$) is the line digraph of dB(q, ℓ). The vertices in dB(q, $\ell + 1$) correspond to edges in dB(q, ℓ). Note that while cycles in dB(q, ℓ) map directly to cycles in dB(q, $\ell + 1$), the converse is not always true: there may be repeated vertices when a cycle from dB(q, $\ell + 1$) is projected down to dB(q, ℓ). The objects in dB(q, ℓ) that correspond to cycles in dB(q, $\ell + 1$) are called *circuits*, which are closed walks with no repeated edges (vertex repetition is allowed).

Proposition 1. *The following are equivalent:*

1. $\mathcal{E}(q, k, \ell) = \frac{q^\ell}{k}$.
2. *There is a partition of the vertex set of* dB(q, ℓ) *into pairwise disjoint* k*-cycles.*
3. *There is a partition of the edge set of* dB(q, $\ell - 1$) *into pairwise edge-disjoint* k*-circuits.*

Proof. (1 $\iff$ 2) Suppose that each vertex of dB(q, ℓ) corresponds to a particular camera view of ℓ consecutive LEDs on some eBug. Rotating the eBug to the left corresponds to following an edge in the graph, since the LEDs shift to the right and one new LED is visible. Hence a cycle of length k in dB(q, ℓ) corresponds to the colouring of a single eBug with k LEDs. A set of multiple disjoint cycles gives an ℓ-valid colouring of multiple eBugs (because vertices are not repeated, each orientation is uniquely identifiable), so the eBug number $\mathcal{E}(q, k, \ell)$ equals the maximum number of *disjoint* k*-cycles* in dB(q, ℓ). If every vertex is in one of the k-cycles, then each colour sequence appears on some eBug. Conversely, if any given colour sequence can be found on some eBug, then the corresponding vertex is in one of the k-cycles. Thus optimal colourings exist exactly when the whole graph can be partitioned into disjoint k-cycles.

(2 $\iff$ 3) The equivalence follows immediately from the line digraph construction. □

Bryant studied edge decompositions of complete directed graphs with loops [3], which correspond to the first order de Bruijn graphs dB(q, 1). The main result of [3] was that dB(q, 1) can be decomposed into k-circuits if and only if $k \geq 3$ and k divides q^2. By Proposition 1, this solves Conjecture 1 for $\ell = 2$.

A similar problem was also posed by Dudeney in 1917 [11], now commonly known as "Dudeney's round table problem". This problem is equivalent to finding a set of Hamiltonian cycles in the complete graph K_n, such that every path of two edges appears in exactly one of the cycles. Dudeney's problem was solved for even n [16], and also some other cases such as when $n - 1$ is a prime power [20]. A generalisation of Dudeney's problem was studied in [17]; here K_n is covered by k-cycles with the same property (instead of n-cycles). The main difference between these problems and Problem 1 is that we are concerned with directed circuits in digraphs, and that we allow loops on the vertices.

There is a body of research on cycle decompositions of complete graphs (see [4] for an introduction and [5] for recent results), and also some work relating to decompositions into fixed-length directed cycles [1]. The methods used, however, are very specific to the special structure of complete graphs, and cannot be applied to de Bruijn graphs. There are also results about decomposing de Bruijn graphs into *variable-length* cycles, using techniques like splitting and merging existing cycles [7]. Golomb's conjecture, which was proven by Mykkeltveit [19], states that the decomposition of binary de Bruijn graphs into the largest number of disjoint cycles is the decomposition into *necklaces* (see Section 4 for a definition). These results, unfortunately, cannot easily be applied to help with Conjecture 1, since the specific requirement of fixed-length cycles is quite restrictive.

2.2. de Bruijn sequences

Note that in the de Bruijn graph dB(q, ℓ), every vertex has in-degree and out-degree q. Also, a path can be found from any vertex u to any vertex v by shifting in letters of v one at a time, so the graph is connected. Hence dB(q, ℓ) is Eulerian, and the next de Bruijn graph dB(q, $\ell + 1$) is Hamiltonian (since an Eulerian circuit in dB(q, ℓ) is equivalent to a Hamiltonian cycle

in dB$(q, \ell+1)$). This simple fact gives us a starting point for Conjecture 1 in the $k = q^\ell$ case: it shows that $\mathcal{E}(q, q^\ell, \ell) = 1$ for every q and ℓ.

Hamiltonian cycles in de Bruijn graphs are called *de Bruijn sequences*. The number of (q, ℓ)-de Bruijn sequences is

$$\frac{(q!)^{q^{\ell-1}}}{q^\ell}.$$

This result is due to van Aardenne-Ehrenfest and de Bruijn [24], and uses an equivalence between spanning arborescences and Eulerian circuits in Eulerian digraphs.

There are several known methods for generating de Bruijn sequences. One construction [21, Section 7.3] gives the lexicographically smallest sequence for any given values of q and ℓ through clever concatenation of necklaces. This method is described in Section 4.1. Another construction involves calculations in finite fields [21, Section 7.7]. This only works when q is a prime power, but has a very simple implementation, which is described in Section 3.1.

3. Linear feedback shift registers

In this section, algebraic properties of finite fields are exploited to find interesting structures in de Bruijn graphs. Section 3.1 describes a well-known construction of de Bruijn sequences; we extend this construction further in Sections 3.2 and 3.3 to find multiple k-cycles in de Bruijn graphs. We assume that the reader is familiar with elementary group and field theory; see [13] for example.

3.1. Construction

Let q be a prime power, and choose a primitive element α from the finite field $F := \mathrm{GF}(q^\ell)$. That is, α generates the multiplicative group $F^* = F \setminus \{0\}$. We may consider F to be an ℓ-dimensional vector space over GF(q), in which case $\{1, \alpha, \alpha^2, \ldots, \alpha^{\ell-1}\}$ is a basis. In particular, α^ℓ can be written as a linear combination of these basis vectors: $\alpha^\ell = p_0 + p_1\alpha + \cdots + p_{\ell-1}\alpha^{\ell-1}$ (this is called the *minimal polynomial* of α over GF(q)).

A *linear feedback shift register* (LFSR) is a digital circuit that generates elements of F^* by successive multiplication by α. The simplest implementation, a *Galois LFSR*, represents the field elements as vectors in $\mathrm{GF}(q)^\ell$ with respect to the basis $\{1, \alpha, \alpha^2, \ldots, \alpha^{\ell-1}\}$. Multiplication of a vector $\boldsymbol{v} := (v_0, v_1 \ldots, v_{\ell-1})$ by α is simply a shift of the vector to the right, except that an α^ℓ term is produced. But α^ℓ can be rewritten in terms of the basis vectors, so the multiplication corresponds to the function $\boldsymbol{v} \mapsto (0, v_0, v_1, \ldots, v_{\ell-2}) + v_{\ell-1}(p_0, p_1, \ldots, p_{\ell-1})$. Since the new state is a linear transformation of the previous state, this function can be expressed as the matrix equation $\boldsymbol{v} \mapsto M\boldsymbol{v}$ (over GF(q)), where the *state change matrix*[1] M is given by

$$M = \begin{pmatrix} 0 & 0 & \cdots & 0 & p_0 \\ 1 & 0 & \cdots & 0 & p_1 \\ 0 & 1 & \cdots & 0 & p_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & p_{\ell-1} \end{pmatrix}.$$

The constants p_i depend on α, and are called the *feedback coefficients* for the LFSR. Note that since F^* is generated by α, repeatedly applying this operation to some non-zero initial vector generates every non-zero vector in $\mathrm{GF}(q)^\ell$.

A *Fibonacci LFSR* is a similar construction that uses the transposed state change matrix M^T. In this configuration, the next state is given by $\boldsymbol{v} \mapsto (v_1, v_2, \ldots, v_{\ell-1}, \sum_{i=0}^{\ell-1} p_i v_i)$. In fact, a Fibonacci LFSR performs the same operation as the corresponding Galois LFSR when the vectors are represented in a different basis. To see this, we must find a matrix C that satisfies $CM = M^T C$.

Let C be defined by $C_{ij} = (M^i)_{0j}$ for $0 \le i < \ell$ and $0 \le j < \ell$ (that is, the (i, j)th entry of C is the $(0, j)$th entry of M^i). The entries of powers of a companion matrix are explicitly known [6], so we can observe that C is a symmetric matrix. Similarly, since $(CM)_{ij} = (M^{i+1})_{0j}$, CM is also symmetric. Hence $CM = (CM)^T = M^T C^T = M^T C$, so C is a change of basis matrix from M to M^T. Note that the first row of C is $\begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix}$, so the first basis vector for the Fibonacci LFSR is also $\alpha^0 = 1$ (as in the Galois LFSR).

We now show that the Fibonacci LFSR follows edges in the corresponding de Bruijn graph. From here onwards, we do not use Galois LFSRs and instead only represent field elements in the Fibonacci basis.

Proposition 2. *Let $F := \mathrm{GF}(q^\ell)$ and fix a primitive $\alpha \in F^*$. If the elements of F are identified with the vertices of* dB(q, ℓ) *by expressing them in the Fibonacci basis over GF(q), then $(\beta, \alpha\beta)$ is an edge in* dB(q, ℓ) *for each $\beta \in F$.*

[1] In linear algebra, this matrix is also known as the *companion matrix* for the minimal polynomial of α.

Proof. Observe that the state change operation $\beta \mapsto \alpha\beta$ in a Fibonacci LFSR corresponds to a left shift of the state vector and an extra term on the end. This is exactly what is required for an edge in $\mathrm{dB}(q, \ell)$. □

Proposition 2 can be used to describe all edges of $\mathrm{dB}(q, \ell)$ in terms of field operations.

Lemma 1. *If $\beta \in F$ and $e \in \mathrm{GF}(q)$, then $(\beta + e, \alpha\beta)$ is an edge in $\mathrm{dB}(q, \ell)$.*

Proof. By Proposition 2, $(\beta, \alpha\beta)$ is an edge in $\mathrm{dB}(q, \ell)$. Recall that in the construction of the Fibonacci basis, the first (or leftmost) component corresponds to the basis vector 1 (the multiplicative identity of F). Thus adding a scalar e to a vector only changes the first component, which is shifted out when following an edge in $\mathrm{dB}(q, \ell)$. Hence β and $\beta + e$ have the same out-neighbours (including $\alpha\beta$). □

Now consider repeatedly applying the state change operation $\beta \mapsto \alpha\beta$ to some initial non-zero field element (1 for example). Since α generates F^*, the Fibonacci LFSR traverses a cycle of length $q^\ell - 1$ in $\mathrm{dB}(q, \ell)$. The missing vertex is 0, and can always be inserted into this cycle by replacing the edge $(1, \alpha)$ with two edges $(1, 0)$ and $(0, \alpha)$. Note that $(1, 0)$ and $(0, \alpha)$ are edges by Lemma 1 (with $\beta = 1, e = -1$ and $\beta = 0, e = 1$, respectively). Thus we have found a Hamiltonian cycle in $\mathrm{dB}(q, \ell)$, which is a de Bruijn sequence.

3.2. Splitting LFSR sequences

Due to the inherently algebraic construction of linear feedback shift registers, the symmetry properties of such sequences may be exploited to produce many cycles of the same length. For this section, we identify vertices of $\mathrm{dB}(q, \ell)$ with the elements of $F = \mathrm{GF}(q^\ell)$ via the Fibonacci basis described above (with respect to a fixed primitive $\alpha \in F^*$).

Fix a value of $k < q^\ell$, and let $\beta_e := \frac{\alpha e}{\alpha^k - 1}$ for each non-zero scalar $e \in \mathrm{GF}(q)^*$. Since $\alpha^{-1}\beta_e + e = \alpha^{k-1}\beta_e$, there is an edge from $\alpha^{k-1}\beta_e$ to β_e in $\mathrm{dB}(q, \ell)$ (by Lemma 1 with $\beta = \alpha^{-1}\beta_e$). We also have $k-1$ other edges $(\alpha^i\beta_e, \alpha^{i+1}\beta_e)$ for $0 \le i < k-1$, so we can form a k-cycle $C_e = (\beta_e, \alpha\beta_e, \ldots, \alpha^{k-1}\beta_e)$ for each of the $q - 1$ values of $e \in \mathrm{GF}(q)^*$. In general, these $q - 1$ cycles are not necessarily pairwise disjoint, but we now show that they are if k is small.

Theorem 4. *$\mathcal{E}(q, k, \ell) \ge q - 1$ for every prime power q and every $k \le m := \frac{q^\ell - 1}{q - 1}$.*

Proof. Let $\log_\alpha \beta \in \mathbb{Z}_{q^\ell - 1}$ be the value of i for which $\alpha^i = \beta$; that is, the discrete logarithm of β with base α. This is well-defined on all of F^* because α is a generator. Note that $\log_\alpha \beta$ is also the position of β in the LFSR sequence (if the initial state is 1). Now consider the relative position of the starting points of two different cycles C_x and C_y as described above, with $x, y \in \mathrm{GF}(q)^*$. The distance along the LFSR sequence between these starting points is

$$\log_\alpha \beta_x - \log_\alpha \beta_y = \log_\alpha \left(\frac{\alpha x}{\alpha^k - 1}\right) - \log_\alpha \left(\frac{\alpha y}{\alpha^k - 1}\right) = \log_\alpha \left(\frac{x}{y}\right).$$

Note that $\frac{x}{y} \in \mathrm{GF}(q)^*$, which is a subgroup of F^* of order $q - 1$. Also note that $(\alpha^m)^{q-1} = 1$, and that $im < q^\ell - 1 = |F^*|$ for $i < q - 1$. Thus α^m has order $q - 1$. But there is only one subgroup of F^* of order $q - 1$ (since F^* is cyclic), so $\frac{x}{y} \in \langle \alpha^m \rangle$. Hence $\frac{x}{y} = (\alpha^m)^j$ for some j, so $\log_\alpha(\frac{x}{y}) = jm$ is an integer multiple of m. Since $k \le m$, the k consecutive vertices of C_x cannot be in C_y, whose starting vertex is at least m places past the start of C_x. Hence these $q - 1$ k-cycles are pairwise disjoint. □

3.3. Translating LFSRs

Fix a scalar $e \in \mathrm{GF}(q)$, and let $\xi_e(\beta) := \alpha\beta + \alpha e$. Note that ξ_e has exactly one fixed point, namely $\varphi_e = \frac{\alpha e}{1 - \alpha}$. Hence

$$\xi_e(\beta + \varphi_e) = \alpha(\beta + \varphi_e) + \alpha e = \alpha\beta + \xi_e(\varphi_e) = \xi_0(\beta) + \varphi_e. \tag{2}$$

By Lemma 1, the q out-neighbours of a vertex β are $\{\xi_e(\beta) \mid e \in \mathrm{GF}(q)\}$. Thus we may partition the edges of $\mathrm{dB}(q, \ell)$ into the q parts $P_e := \{(\beta, \xi_e(\beta)) \mid \beta \in \mathrm{GF}(q^\ell)\}$, where $e \in \mathrm{GF}(q)$. Note that (2) ensures that $(x, y) \in P_0$ implies $(x + \varphi_e, y + \varphi_e) \in P_e$. Hence if $(\beta_1, \beta_2, \ldots, \beta_k, \beta_1)$ is a circuit contained in P_0, then $(\beta_1 + \varphi_e, \beta_2 + \varphi_e, \ldots, \beta_k + \varphi_e, \beta_1 + \varphi_e)$ is a circuit contained in P_e. We call this operation *translating* the circuit by e. Note that the q translations of a circuit contained in P_0 are pairwise edge-disjoint because the P_e are disjoint.

We are now ready to prove Theorem 1, which we restate here:

Theorem 1. *If q is a prime power and $\ell \ge 1$, then $\mathcal{E}(q, q^{\ell-1}, \ell) = q$.*

Proof. If $\ell = 1$, then the result is trivial since $\mathrm{dB}(q, 1)$ contains q loops. Hence we may assume that $\ell \ge 2$.

Let $k = q^{\ell-1}$. By Proposition 1, it is sufficient to find a partition of $\mathrm{dB}(q, \ell - 1)$ into q edge-disjoint k-circuits.

Recall from Section 3.1 that the LFSR sequence is constructed using edges solely of the form $(\beta, \alpha\beta) \in P_0$, and forms a cycle C_0 of length $k-1$. This cycle (of vertices) is also a circuit of $k-1$ edges, and we can construct a translated $(k-1)$-circuit C_e in P_e for each scalar $e \in \mathrm{GF}(q)$.

The circuit C_0 contains every edge from P_0 except the loop $(0, 0) = (\varphi_0, \varphi_0)$, which translates to another loop (φ_e, φ_e) in P_e. Since C_0 contains every non-zero vertex, we may insert, say, the loop (φ_1, φ_1) into C_0 to obtain a k-circuit $\widehat{C}_0$ (note that $\widehat{C}_0$ is no longer a cycle since it contains the vertex φ_1 twice). Similarly, we may insert the loop $(\varphi_{e+1}, \varphi_{e+1})$ into C_e to generate a k-circuit $\widehat{C}_e$ for each $e \in \mathrm{GF}(q)$.

Observe that each edge $(\beta, \xi_e(\beta))$ of $\mathrm{dB}(q, \ell - 1)$ appears in a unique circuit: if $\beta = \varphi_e$, then the edge is a loop in $\widehat{C}_{e-1}$; otherwise it is in $\widehat{C}_e$. Thus we have a partition of $\mathrm{dB}(q, \ell - 1)$ into q edge-disjoint k-circuits $\{\widehat{C}_e \mid e \in \mathrm{GF}(q)\}$. Hence $\mathcal{E}(q, q^{\ell-1}, \ell) \geq q$. □

For example, we may apply Theorem 1 with $q = 3$ and $\ell = 4$ to obtain three cycles of length 9. Suppose we choose a primitive $\alpha \in F = \mathrm{GF}(3^3)$ whose minimal polynomial over GF(3) is $\alpha^3 = 2 + \alpha$. We now construct a 26-cycle C_0 in dB(3, 3) by iterating the LFSR starting from vertex 100, which corresponds to the field element 1 (the multiplicative identity of F). This produces the following cycle of symbols: $C_0 = 10020212210222001012112011$ (the corresponding cycle of vertices is 100, 001, 002, . . . , 011, 111, 110).

Now we may construct the 27-circuit $\widehat{C}_0$ by inserting a loop, say (111, 111), into C_0. The three translations of $\widehat{C}_0$, shown below in (3), partition the edge set of dB(3, 3), and hence the corresponding cycles in dB(3, 4) partition the vertex set of dB(3, 4). Thus we have shown that $\mathcal{E}(3, 27, 4) = 3$.

$$\begin{aligned}
\widehat{C}_0 &= 100202122102220010121120111,\\
\widehat{C}_1 &= 211010200210001121202201222,\\
\widehat{C}_2 &= 022121011021112202010012000.
\end{aligned} \tag{3}$$

3.4. LFSRs from non-primitive elements

Suppose that in the construction of the LFSR, we chose a non-primitive element β with multiplicative order $k < q^\ell - 1$. If $\{1, \beta, \beta^2, \ldots, \beta^{\ell-1}\}$ is still a basis of $F = \mathrm{GF}(q^\ell)$ over GF(q), then vectors with respect to this basis are still in correspondence with field elements. Repeated multiplication by β, however, no longer generates every element of F^*; instead this process traverses the cyclic subgroup of order k generated by β. Thus the action of the LFSR traces out this subgroup of F^* if the initial state is the identity 1. This corresponds to a k-cycle in $\mathrm{dB}(q, \ell)$.

Choosing a different starting state for the LFSR translates the whole sequence, but does not change the length of the cycle. This gives a partition of the non-zero vertices into k-cycles. The number of these cycles is $|F^*/\langle\beta\rangle| = \frac{q^\ell - 1}{k}$, giving $\mathcal{E}(q, k, \ell) \geq \frac{q^\ell - 1}{k}$.

Theorem 5. *Let q be a prime power, and k a factor of $q^\ell - 1$. If k does not divide $q^i - 1$ for each $i < \ell$, then*

$$\mathcal{E}(q, k, \ell) = \frac{q^\ell - 1}{k}.$$

Proof. Since k divides $q^\ell - 1$, there is an element β of multiplicative order k in $F = \mathrm{GF}(q^\ell)$. Since k does not divide $q^i - 1$ for $i < \ell$, β is not in any subfield $\mathrm{GF}(q^i)$ of F. Hence β is not the root of any polynomial over GF(q) of degree $d < \ell$. Therefore $\{1, \beta, \beta^2, \ldots, \beta^{\ell-1}\}$ is linearly independent over GF(q), and hence a basis of F.

Therefore, the LFSR generated by β traces out a distinct k-cycle in $\mathrm{dB}(q, \ell)$ for each equivalence class of $F^*/\langle\beta\rangle$. We have found $\frac{q^\ell - 1}{k}$ disjoint k-cycles in $\mathrm{dB}(q, \ell)$. Since the k-cycles cover all but one vertex in $\mathrm{dB}(q, \ell)$ and $k > 1$, this is the best possible bound. □

By Zsigmondy's theorem [30], there is a prime p that divides $q^\ell - 1$ but not $q^i - 1$ for $i < \ell$ for any q and ℓ, except when $(q, \ell) = (2, 6)$ or $\ell = 2$ and q is a Mersenne prime (that is, $q = 2^{p'} - 1$ for some prime p'). Thus Theorem 5 can be applied with $k = p$ to obtain an almost optimal eBug colouring with p LEDs on each eBug (only one colour sequence is unused). Furthermore, if larger eBugs are desired for the same values of q and ℓ, any multiple of p that divides $q^\ell - 1$ can also be used for k.

4. Necklaces

This section focuses on the combinatorics of words to find and combine cycles in a de Bruijn graph. A *word* of length k over an alphabet A is a sequence of k *letters*, each of which is an element of A. We often use the *left rotation* operation ρ, which cyclically permutes the order of letters in a word: $\rho(a_1a_2 \ldots a_k) := a_2a_3 \ldots a_ka_1$. We may rotate a word by any amount by repeatedly applying ρ; ρ^i rotates a word by i places to the left.

Usually, a *factor* f of a word w is defined as any block of consecutive letters in w, and f is a *prefix* if it appears at the start of w. In this case, if w has length k, there are at most $k - \ell$ factors of length ℓ (or ℓ-factors) of w. For this section, we allow factors to "wrap around", so that f is a factor of w if and only if it is a prefix of some rotation $\rho^i(w)$. This way, it is possible to have k different ℓ-factors of a word of length k.

4.1. Necklaces and de Bruijn graphs

Definition 3. A q-ary *necklace* is an equivalence class of words over the alphabet $\mathbb{Z}_q$ under cyclic rotation ρ^i. The *length* of a necklace is the length of any word in the class, while the *size* of a necklace is the number of words in the class. A necklace with equal length and size is called *aperiodic*.

Every word in a q-ary length ℓ necklace corresponds to a vertex in the de Bruijn graph $\mathrm{dB}(q, \ell)$, and a cyclic rotation corresponds to following an edge in this graph. Thus a size t necklace can be thought of as a t-cycle in $\mathrm{dB}(q, \ell)$. Note that every vertex is part of some necklace, so the vertex set of $\mathrm{dB}(q, \ell)$ can be partitioned into necklaces.

For a fixed q and ℓ, the possible necklace sizes in $\mathrm{dB}(q, \ell)$ are the divisors of ℓ. Moreau's necklace counting function [18], shown below in (4), gives the number of q-ary size t necklaces, and is defined in terms of the Möbius function μ.

$$M(q, t) = \frac{1}{t} \sum_{d|t} \mu\left(\frac{t}{d}\right) q^d. \tag{4}$$

The total number of length ℓ necklaces is more easily calculated using Euler's totient function, φ:

$$Z(q, \ell) = \sum_{t|\ell} M(q, t) = \frac{1}{\ell} \sum_{d|\ell} \varphi\left(\frac{\ell}{d}\right) q^d.$$

When ℓ is prime, there are exactly q necklaces of size 1 (the constant words); the remainder of the necklaces have size ℓ. Thus there are $\frac{q^\ell - q}{\ell}$ disjoint ℓ-cycles in $\mathrm{dB}(q, \ell)$. Hence $\mathcal{E}(q, \ell, \ell) \geq \frac{q^\ell - q}{\ell}$ for any prime ℓ. Note that when $\ell > q$, this lower bound is tight since there are less than ℓ remaining vertices in $\mathrm{dB}(q, \ell)$.

A *Lyndon word* is the lexicographically smallest representative of an aperiodic necklace. It is possible to construct a de Bruijn sequence for $\mathrm{dB}(q, \ell)$ by concatenating all q-ary Lyndon words whose length divides ℓ in lexicographic order. In fact, the sequence that is generated is the lexicographically smallest de Bruijn sequence of the given order [14].

4.2. Multiplying necklaces

Suppose we have two systems of coloured eBugs, where each colouring is ℓ-valid. In this section, we describe a type of direct product that yields many more eBugs at the expense of using more colours. The result is summarised in Theorem 2.

Instead of modelling the colouring problem with de Bruijn graphs, we find a set of necklaces of length k that correspond to the disjoint k-cycles in $\mathrm{dB}(q, \ell)$. The definition of an ℓ-*valid* set of necklaces translates directly from Definition 1.

Theorem 2. *Fix a value of ℓ and set $\mathcal{E}_1 = \mathcal{E}(q_1, k_1, \ell)$ and $\mathcal{E}_2 = \mathcal{E}(q_2, k_2, \ell)$. Then*

$$\mathcal{E}(q_1 q_2, \mathrm{lcm}(k_1, k_2), \ell) \geq \gcd(k_1, k_2)\, \mathcal{E}_1\, \mathcal{E}_2.$$

Proof. We first demonstrate this proof for the special case of $k_1 = k_2 = k$ (so $\gcd(k_1, k_2) = \mathrm{lcm}(k_1, k_2) = k$), and then show that the construction can be extended to the general case. The construction describes a one-to-k mapping from pairs of necklaces to necklaces with $q_1 q_2$ colours.

In order to construct necklaces over a larger alphabet, we use pairs of letters (colours) as the letters in the resulting necklaces. We define a *merging* operation $\mathcal{M}$ that pairs corresponding letters from two words of the same length: if $a := a_1 a_2 \ldots a_k$ and $b := b_1 b_2 \ldots b_k$, then $\mathcal{M}(a, b) = (a_1, b_1)(a_2, b_2) \ldots (a_k, b_k)$.

Let N_i be an ℓ-valid set of $\mathcal{E}_i$ q_i-ary necklaces of length k, for $i = 1, 2$. For each necklace $n \in N_1 \cup N_2$, choose a representative word w_n. Now, for a pair of necklaces $(n_1, n_2) \in N_1 \times N_2$, we construct k new words $\mathcal{M}(w_{n_1}, \rho^i(w_{n_2}))$ over $\mathbb{Z}_{q_1} \times \mathbb{Z}_{q_2}$, where $i \in \mathbb{Z}_k$. Note that we only rotate one of the words, since rotating both by the same amount creates a word that is equivalent under cyclic rotation (it is only the relative rotation that matters). An example of this process with $k = 8$, $\ell = 3$ and $q_1 = q_2 = 2$ is illustrated in Fig. 4.

This process can be performed for every pair $(n_1, n_2) \in N_1 \times N_2$, generating k new words every time. Hence it is clear that $k\mathcal{E}_1\mathcal{E}_2$ words are produced, and that $q_1 q_2$ colours are used. Thus it remains only to show that the set of corresponding necklaces is ℓ-valid.

In each of the original necklaces $n \in N_1 \cup N_2$, there are k different ℓ-factors (since the N_i were ℓ-valid). Thus the total number of distinct ℓ-factors in N_i is $k\mathcal{E}_i$. Suppose the word $a := a_1 a_2 \ldots a_\ell$ occurs in the necklace $n_a \in N_1$, and $b := b_1 b_2 \ldots b_\ell$ occurs in $n_b \in N_2$. There is a unique i such that w_{n_a} and $\rho^i(w_{n_b})$ have a and b aligned, so $\mathcal{M}(w_{n_a}, \rho^i(w_{n_b}))$ must contain the factor $(a_1, b_1)(a_2, b_2) \ldots (a_\ell, b_\ell)$. Since there are $k\mathcal{E}_1 \times k\mathcal{E}_2$ pairs of ℓ-factors from the original necklaces, there must be at least $k\mathcal{E}_1 \times k\mathcal{E}_2$ distinct ℓ-factors in the set of merged words. But there are only k possible ℓ-factors in each of the $k\mathcal{E}_1\mathcal{E}_2$ merged words, so each ℓ-factor must appear exactly once. Therefore the set of necklaces corresponding to the merged words is ℓ-valid.

To generalise to the case where $k_1 \neq k_2$, we can traverse the original necklaces multiple times to obtain words of length $\mathrm{lcm}(k_1, k_2)$. For a necklace $n \in N_i$, pick a representative word of n and repeat it $\frac{\mathrm{lcm}(k_1, k_2)}{k_i}$ times to obtain w_n. This way we may still merge words from N_1 and N_2 using $\mathcal{M}$ (since they are the same length).

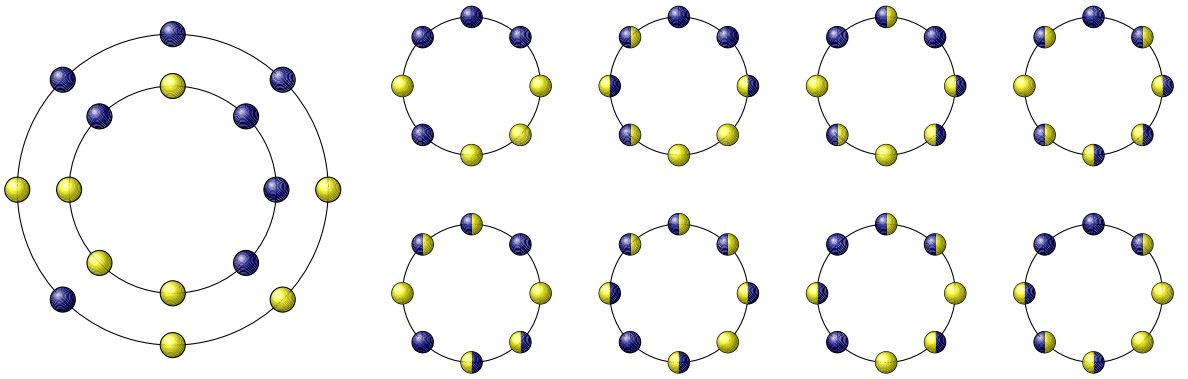


Fig. 4. Two necklaces being multiplied to produce many new necklaces. Ordered pairs of colours are used to specify the new colours in the resulting necklaces.

To model rotations of w_{n_1} and w_{n_2}, we act on the pair with elements of the group $\mathbb{Z}_{k_1} \times \mathbb{Z}_{k_2}$. Simultaneous rotation of w_{n_1} and w_{n_2} (by the same amount) only rotates the merged word that is produced, so we may identify unique merged words with elements of the quotient group $R \coloneqq \mathbb{Z}_{k_1} \times \mathbb{Z}_{k_2}/\langle(1,1)\rangle$. Hence each pair (w_{n_1}, w_{n_2}) produces $|R| = \frac{k_1 k_2}{\operatorname{lcm}(k_1,k_2)} = \gcd(k_1, k_2)$ unique words, for a total of $\gcd(k_1, k_2)\,\mathcal{E}_1\mathcal{E}_2$ words.

It remains to show that the set of merged words is ℓ-valid. The number of ℓ-factors that appear in N_i is $k_i\mathcal{E}_i$. As before, suppose the ℓ-factor a occurs in $n_a \in N_1$ and b occurs in $n_b \in N_2$. Define i and j so that a is the ℓ-prefix of $\rho^i(w_{n_a})$, and b is the ℓ-prefix of $\rho^j(w_{n_b})$. Note that rotating both words together keeps a and b aligned, so the pair (i, j) corresponds to a unique element of R, and thus $\mathcal{M}(a, b)$ appears in one of the merged words. Hence there are at least $k_1\mathcal{E}_1 \times k_2\mathcal{E}_2$ distinct ℓ-factors in the set of merged words. The number of possible positions for these ℓ-factors is the product of the number of merged words and the length of each word. This number is $\gcd(k_1, k_2)\,\mathcal{E}_1\mathcal{E}_2 \times \operatorname{lcm}(k_1, k_2) = k_1k_2\mathcal{E}_1\mathcal{E}_2$, so all ℓ-factors are unique, and the set of corresponding necklaces is ℓ-valid. Hence $\mathcal{E}(q_1q_2, \operatorname{lcm}(k_1, k_2), \ell) \geq \gcd(k_1, k_2)\,\mathcal{E}_1\,\mathcal{E}_2$. □

The conditions in Theorem 2 guarantee that if the original colourings are optimal, then the resulting colouring is also optimal. This allows a result for prime powers, such as Theorem 1, to be extended to any integer by repeated application of Theorem 2 (after applying Theorem 1 for each prime power factor in the prime decomposition of q).

Corollary 1. *$\mathcal{E}(q, q^{\ell-1}, \ell) = q$ for all q and ℓ.*

On the other hand, if the original colourings are not optimal, the product colouring may be even "less" optimal. For example, the construction in Section 3.4 is almost optimal since it uses all but one vertex of $\mathrm{dB}(q_1, \ell)$. If we use Theorem 2 to multiply this with an optimal colouring (where the k-cycles use every vertex of $\mathrm{dB}(q_2, \ell)$), there would be q_2^ℓ vertices of $\mathrm{dB}(q_1q_2, \ell)$ not used by a cycle in the product colouring.

4.3. Interleaving necklaces

Theorem 3. *$\mathcal{E}(q, tk, t\ell) \geq \frac{k^{t-1}}{t}\mathcal{E}(q, k, \ell)^t$ whenever t divides k.*

Proof. Let N be an ℓ-valid set of $\mathcal{E}(q, k, \ell)$ necklaces of length k over the alphabet $\mathbb{Z}_q$. For each necklace $n \in N$, fix a specific representative word w_n. We now construct the set $W \subseteq \mathbb{Z}_q^k$ of all words that appear in some necklace from N:

$$W \coloneqq \{\rho^i(w_n) \mid n \in N,\ i \in \mathbb{Z}_k\}.$$

Since N is ℓ-valid, the necklaces of N are aperiodic. Thus we may uniquely identify how far each word in W is rotated from its representative w_n. For each word $w = \rho^i(w_n) \in W$, define $\psi(w) \coloneqq i \in \mathbb{Z}_k$.

We now define a function $\mathcal{I}$ that interleaves the letters of multiple words to construct a single long word.

$$\mathcal{I}(a_{11}a_{12}\ldots a_{1k}, a_{21}a_{22}\ldots a_{2k}, \ldots, a_{t1}a_{t2}\ldots a_{tk}) \coloneqq a_{11}a_{21}\ldots a_{t1}a_{12}a_{22}\ldots a_{t2}a_{13}\ldots a_{tk}.$$

Note that $\mathcal{I}$ is injective, since we can deinterleave the resulting word to recover the original words. We write $\mathcal{I}^{-1}$ for this deinterleaving function, which gives a t-tuple of k words from a single word of length tk. For brevity, we write t-tuples of words in boldface as $\boldsymbol{w} \coloneqq (w_1, w_2, \ldots, w_t) \in W^t$. We also extend ψ to operate on t-tuples, so we may write $\psi(\boldsymbol{w}) \coloneqq (\psi(w_1), \psi(w_2), \ldots, \psi(w_t))$.

Now consider the set $V \coloneqq \{\mathcal{I}(\boldsymbol{w}) \mid \boldsymbol{w} \in W^t, \psi(w_1) = 0, \sum_i \psi(w_i) \equiv 0 \bmod t\}$. Note that the $\psi(w_i)$ are in $\mathbb{Z}_k$ and t divides k, so taking the sum modulo t is well-defined. Simple arithmetic shows that $|V| = \frac{|W|^t}{kt} = \frac{k^{t-1}}{t}|N|^t$. We claim that V is a $t\ell$-valid set of words.

Take any $v = \mathcal{I}(\boldsymbol{w})$ from V, and observe the following property of the interleaving function:

$$\rho(v) = \mathcal{I}(w_2, w_3, \ldots, w_t, \rho(w_1)). \tag{5}$$

Thus if $\psi(\boldsymbol{w}) = (z_1, z_2, \ldots, z_t)$, then $\psi(\mathcal{I}^{-1}(\rho(v))) = (z_2, z_3, \ldots, z_t, z_1 + 1)$. Hence rotating v to the left increases $\sum \psi(w_i)$ by 1. By iterating (5), we can find all tk rotations of v. Of these tk rotations, only k have $\sum \psi(w_i) \equiv 0 \bmod t$ (every tth rotation), and only one of these k has the first word not rotated ($\psi(w_1) = 0$). Hence no other $v' \in V$ is a rotation of v.

Now suppose there are $v, v' \in V$ that share a $t\ell$-factor u (in any position). Rotate v and v', respectively, to find $\rho^z(v)$ and $\rho^{z'}(v')$, both of which have u as their $t\ell$-prefix. Note that $\rho^z(v)$ and $\rho^{z'}(v')$ are not necessarily in V, but are still obtained by interleaving a t-tuple of words from W (see (5)). We may deinterleave u into a t-tuple of length ℓ words $\mathcal{I}^{-1}(u) = \boldsymbol{u} := (u_1, u_2, \ldots, u_t)$. Each of these words u_i is the ℓ-prefix of a unique word $w_i \in W$ due to the ℓ-validity of N. Thus the only interleaved word with u as its $t\ell$-prefix is $\mathcal{I}(\boldsymbol{w})$, and hence $\rho^z(v) = \mathcal{I}(\boldsymbol{w}) = \rho^{z'}(v')$. But v' is not a rotation of v, so $v = v'$. Therefore, V is a $t\ell$-valid set of length tk words and $\mathcal{E}(q, tk, t\ell) \geq |V| = \frac{k^{t-1}}{t}\mathcal{E}(q, k, \ell)^t$. □

As an example, we now apply Theorem 3 with $q = 4$, $k = 8$, $\ell = 3$ and $t = 2$ to the eight necklaces of length 8 in Fig. 4, which in turn were constructed using Theorem 2. These necklaces may be more compactly written as strings: 00030333, 10021233, 11020323, 11120232, 01130223, 10131222, 01031322, 00121332. Suppose we wish to interleave the first and second necklaces in all allowable rotations. We must keep the first necklace fixed and only rotate the second necklace by even amounts to satisfy the conditions of the set V:

00030333	00030333	00030333	00030333
10021233	02123310	12331002	33100212.

We may now interleave these necklaces to obtain the following four necklaces of length 16:

0100003201323333 0002013203333130 0102033301303032 0303013000323132.

But this is for just one pair of necklaces; we can repeat this procedure for every ordered pair of necklaces from our original list of eight. For each of the 64 possible pairs, we produce four new necklaces of length 16, yielding a total of 256 necklaces. These are listed in Table 1, with each line corresponding to a particular pair of necklaces. Hence $\mathcal{E}(4, 16, 6) = 256$.

As with Theorem 2, if the original colouring is optimal, then the colouring obtained by interleaving the necklaces is also optimal. This allows us to extend existing results by recursively applying Theorem 3.

Corollary 2. *If every prime factor of t divides q, then*

$$\mathcal{E}(q, tq^{\ell}, t\ell) = \frac{q^{(t-1)\ell}}{t}, \quad \textit{and}$$
$$\mathcal{E}(q, tq^{\ell-1}, t\ell) = \frac{q^{(t-1)\ell+1}}{t}.$$

Proof. Recall that $\mathcal{E}(q, q^{\ell}, \ell) = 1$ and $\mathcal{E}(q, q^{\ell-1}, \ell) = q$ (see Corollary 1). Now apply Theorem 3 repeatedly using each prime factor of t. Note that since each prime factor of t divides q, and each value of k in this process is a multiple of q, Theorem 3 is applicable at each step. □

We also have a partial extension of Theorem 3, which allows us to interleave a pair of necklaces of odd length:

Theorem 6. $\mathcal{E}(q, 2k, 2\ell) \geq \left\lfloor \frac{k}{2} \right\rfloor \mathcal{E}(q, k, \ell)^2$.

The proof of Theorem 6 is analogous to that of Theorem 3, except that a slightly different condition is used in the construction of the set V. Here, $V := \{\mathcal{I}(w_1, w_2) \mid w_1, w_2 \in W, \psi(w_1) = 0, \psi(w_2) < \frac{k-1}{2}\}$.

4.4. Necklace concatenation

Whenever two length ℓ necklaces share an $(\ell - 1)$-factor, the corresponding cycles in $\mathrm{dB}(q, \ell)$ can be concatenated. This is because the corresponding edge circuits in $\mathrm{dB}(q, \ell - 1)$ have a common vertex, and can thus be joined to create a larger circuit, which in turn gives a larger cycle in $\mathrm{dB}(q, \ell)$. This relationship between necklaces turns out to be very useful, so we construct a *necklace adjacency graph* $N(q, \ell)$. The q-ary length ℓ necklaces form the vertex set of $N(q, \ell)$, while pairs of necklaces that share an $(\ell - 1)$-factor are joined by an edge.

Consider any (connected) subtree S in $N(q, \ell)$. By applying the above operation for each edge in S, the cycles for each necklace in S can be concatenated together to produce one long cycle, whose length is the sum of the sizes of the individual necklaces. Hence if we find a spanning forest in $N(q, \ell)$ where each component subtree has k as the total size of its necklaces, we can partition $\mathrm{dB}(q, \ell)$ into k-cycles. If the forest does not span $N(q, \ell)$, this still gives a lower bound on the eBug number: if there are m component subtrees in the forest, each with total necklace size k, then $\mathcal{E}(q, k, \ell) \geq m$.

Table 1
Example of interleaved necklaces. Eight 8-cycles from dB(4, 3) were used to produce these 256 16-cycles in dB(4, 6).

(0, 0):	0000003300333333	0003003303333030	0003033300303033	0303003000333033
(0, 1):	0100003201323333	0002013203333130	0102033301303032	0303013000323132
(0, 2):	0101003200333233	0002003302333131	0003023301313032	0203013100323033
(0, 3):	0101013200323332	0102003203323131	0002033201313132	0302013101323032
(0, 4):	0001013300323233	0103003202333031	0002023300313133	0203003101333032
(0, 5):	0100013301323232	0103013202323130	0102023201303133	0202013001333132
(0, 6):	0001003301333232	0003013302323031	0103023200313033	0202003100333133
(0, 7):	0000013201333332	0102013303323030	0103033200303132	0302003001323133
(1, 0):	1000002310233333	1003002313233030	1003032310203033	1303002010233033
(1, 1):	1100002211223333	1002012213233130	1102032311203032	1303012010223132
(1, 2):	1101002210233233	1002002312233131	1003022311213032	1203012110223033
(1, 3):	1101012210223332	1102002213223131	1002032211213132	1302012111223032
(1, 4):	1001012310223233	1103002212233031	1002022310213133	1203002111233032
(1, 5):	1100012311223232	1103012212223130	1102022211203133	1202012011233132
(1, 6):	1001002311233232	1003012312223031	1103022210213033	1202002110233133
(1, 7):	1000012211233332	1102012313223030	1103032210203132	1302002011223133
(2, 0):	1010002300332333	1013002303332030	1013032300302033	1313002000332033
(2, 1):	1110002201322333	1012012203332130	1112032301302032	1313012000322132
(2, 2):	1111002200332233	1012002302332131	1013022301312032	1213012100322033
(2, 3):	1111012200322332	1112002203322131	1012032201312132	1312012101322032
(2, 4):	1011012300322233	1113002202332031	1012022300312133	1213002101332032
(2, 5):	1110012301322232	1113012202322130	1112022201302133	1212012001332132
(2, 6):	1011002301332232	1013012302322031	1113022200312033	1212002100332133
(2, 7):	1010012201332332	1112012303322030	1113032200302132	1312002001322133
(3, 0):	1010102300233323	1013102303233020	1013132300203023	1313102000233023
(3, 1):	1110102201223323	1012112203233120	1112132301203022	1313112000223122
(3, 2):	1111102200233223	1012102302233121	1013122301213022	1213112100223023
(3, 3):	1111112200223322	1112102203223121	1012132201213122	1312112101223022
(3, 4):	1011112300223223	1113102202233021	1012122300213123	1213102101233022
(3, 5):	1110112301223222	1113112202223120	1112122201203123	1212112001233122
(3, 6):	1011102301233222	1013112302223021	1113122200213023	1212102100233123
(3, 7):	1010112201233322	1112112303223020	1113132200203122	1312102001223123
(4, 0):	0010103300232333	0013103303232030	0013133300202033	0313103000232033
(4, 1):	0110103201222333	0012113203232130	0112133301202032	0313113000222132
(4, 2):	0111103200232233	0012103302232131	0013123301212032	0213113100222033
(4, 3):	0111113200222332	0112103203222131	0012133201212132	0312113101222032
(4, 4):	0011113300222233	0113103202232031	0012123300212133	0213103101232032
(4, 5):	0110113301222232	0113113202222130	0112123201202133	0212113001232132
(4, 6):	0011103301232232	0013113302222031	0113123200212033	0212103100232133
(4, 7):	0010113201232332	0112113303222030	0113133200202132	0312103001222133
(5, 0):	1000103310232323	1003103313232020	1003133310202023	1303103010232023
(5, 1):	1100103211222323	1002113213232120	1102133311202022	1303113010222122
(5, 2):	1101103210232223	1002103312232121	1003123311212022	1203113110222023
(5, 3):	1101113210222322	1102103213222121	1002133211212122	1302113111222022
(5, 4):	1001113310222223	1103103212232021	1002123310212123	1203103111232022
(5, 5):	1100113311222222	1103113212222120	1102123211202123	1202113011232122
(5, 6):	1001103311232222	1003113312222021	1103123210212023	1202103110232123
(5, 7):	1000113211232322	1102113313222020	1103133210202122	1302103011222123
(6, 0):	0010003310332323	0013003313332020	0013033310302023	0313003010332023
(6, 1):	0110003211322323	0012013213332120	0112033311302022	0313013010322122
(6, 2):	0111003210332223	0012003312332121	0013023311312022	0213013110322023
(6, 3):	0111013210322322	0112003213322121	0012033211312122	0312013111322022
(6, 4):	0011013310322223	0113003212332021	0012023310312123	0213003111332022
(6, 5):	0110013311322222	0113013212322120	0112023211302123	0212013011332122
(6, 6):	0011003311332222	0013013312322021	0113023210312023	0212003110332123
(6, 7):	0010013211332322	0112013313222020	0113033210302122	0312003011322123
(7, 0):	0000102310333323	0003102313333020	0003132310303023	0303102010333023
(7, 1):	0100102211323323	0002112213333120	0102132311303022	0303112010323122
(7, 2):	0101102210333223	0002102312333121	0003122311313022	0203112110323023
(7, 3):	0101112210323322	0102102213323121	0002132211313122	0302112111323022
(7, 4):	0001112310323223	0103102212333021	0002122310313123	0203102111333022
(7, 5):	0100112311323222	0103112212323120	0102122211303123	0202112011333122
(7, 6):	0001102311333222	0003112312323021	0103122210313023	0202102110333123
(7, 7):	0000112211333322	0102112313323020	0103132210303122	0302102011323123

In the case when ℓ is prime, recall that most necklaces have size ℓ. Let $N'(q, \ell)$ be the subgraph of $N(q, \ell)$ induced by the size ℓ necklaces. Suppose that there is a perfect matching in $N'(q, \ell)$: this is a spanning forest, and each component subtree has total size 2ℓ. Hence concatenating the cycles for each pair of necklaces in the matching produces $\frac{q^\ell - q}{2\ell}$ cycles in dB(q, ℓ), each of length 2ℓ. Similarly, we may generalise this to larger multiples of ℓ:

Proposition 3. *Let ℓ be a prime. If there is a spanning forest of $N'(q, \ell)$ in which each component subtree has t vertices, then $\mathcal{E}(q, t\ell, \ell) \geq \frac{q^\ell - q}{t\ell}$, with equality if $q < t\ell$.*

In particular, the existence of a Hamiltonian path in $N'(q, \ell)$ is sufficient to apply Proposition 3 for any t that divides $\frac{q^\ell - q}{\ell}$ (removing every tth edge from the path produces the required forest). A Hamiltonian path in $N(q, \ell)$ is called a *Gray code* for necklaces, and is conjectured to exist whenever q or ℓ is odd [22] (if both q and ℓ are even, a simple parity argument reveals that $N(q, \ell)$ is bipartite with unequal parts). It is a trivial matter to transform the Hamiltonian path in $N(q, \ell)$ to one in $N'(q, \ell)$, since all neighbours of a constant necklace are adjacent to each other. For $q = 2$, there is also an existing construction of a 2-*Gray code* for fixed density necklaces [25], which lists every necklace with a fixed number of 0s such that consecutive necklaces differ in exactly two places (a 0 and 1 are exchanged).

4.5. Robot identification with necklaces

In some applications of robot identification, orientation information is either not required or can be obtained through different means [26,27]. This relaxes the conditions necessary to obtain a valid colouring of eBugs, since a particular colour sequence may appear more than once on a single eBug. In the de Bruijn graph, this amounts to finding the maximum number of disjoint closed k-walks (instead of k-cycles). When ℓ is a divisor of k, there is an easy solution that turns out to be the best possible; this is a direct corollary of Golomb's conjecture.

Proposition 4. *If ℓ is a divisor of k, then the maximum number of pairwise disjoint closed k-walks in $\mathrm{dB}(q, \ell)$ is $Z(q, \ell)$, the number of q-ary length ℓ necklaces.*

Proof. Golomb's conjecture, which was proved by Mykkeltveit [19], states that the maximum number of pairwise disjoint cycles (of any length) in $\mathrm{dB}(q, \ell)$ is $Z(q, \ell)$. Since each closed walk contains a cycle, there are at most $Z(q, \ell)$ pairwise disjoint closed k-walks in $\mathrm{dB}(q, \ell)$.

Now consider any length ℓ necklace. The size t of this necklace must divide ℓ and hence k, so the corresponding t-cycle in $\mathrm{dB}(q, \ell)$ can be traversed multiple times to obtain a closed k-walk. Thus there are $Z(q, \ell)$ pairwise disjoint closed k-walks in the graph. □

5. Concluding remarks

The theorems presented in Sections 3 and 4 focus on finding large eBug colourings to obtain bounds on the eBug number $\mathcal{E}(q, k, \ell)$. In particular, we concentrated on constructions that yield optimal colourings to support Conjecture 1. The algebraic construction in Theorem 1, and its corresponding extension in Corollary 1, produce q eBugs of maximal size for any q and ℓ. The combinatorial results in Section 4 increase the number of eBugs by moving to a larger de Bruijn graph. These results can even produce eBug colourings for practical applications: for example, we have $\mathcal{E}(2, 16, 5) = 2$ by Theorem 1, so $\mathcal{E}(4, 16, 5) = 64$ by Theorem 2. The current eBugs have 16 LEDs, a camera can distinguish four colours in an image quite easily, and five consecutive LEDs are visible in practice, so it is possible to construct a network of 64 uniquely identifiable eBugs. Also, Theorem 3 can be used to significantly increase the number of eBugs without increasing the number of colours, q. Moreover, it keeps the ratio $\frac{k}{\ell}$ constant, which is a reasonable assumption when designing such a network (since the camera can see a fixed arc of the circle of LEDs).

Unfortunately, the only way to produce many eBugs (more than q) with our results is by applying Theorem 2 or Theorem 3, which necessarily increase either q, the number of colours, or ℓ, the number of consecutive visible LEDs. The major remaining gap appears to be for prime q and ℓ, with $k < q^{\ell-1}$, since the multiplying/interleaving constructions cannot produce them.

Ideally, we would like to be able to have many small eBugs (small number of LEDs) with a small number of colours. For example, we would like to show that $\mathcal{E}(q, q^{\ell-i}, \ell) = q^i$ when ℓ is large enough (for each i). Probabilistic arguments may be useful in finding such colourings, but a constructive approach is preferable for applications to robot networks (the search space becomes too large even for small practical examples, and in many cases the colourings appear to be quite rare).

Solving Problem 1 is much harder, since there are no optimal colourings in the cases not covered by Conjecture 1. Improving the lower bounds in these cases, however, is likely to be a much easier task. We have shown that near-optimal colourings exist in many cases.

Acknowledgements

We would like to thank Heiko Dietrich for introducing us to Zsigmondy's Theorem, and suggesting that it can be used to find examples to which Theorem 5 can be applied. We also thank the referees for pointing out some related references and for their helpful suggestions. The third author's research was supported by the Australian Research Council.

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